Poisoning and Evasion: Deep Learning-Based NIDS under Adversarial Attacks

1st Hesamodin Mohammadian
Canadian Institute for Cybersecurity (CIC)
University of New Brunswick
Fredericton, NB, Canada
h.mohammadian@unb.ca

2nd Arash Habibi Lashkari
Behaviour-Centric Cybersecurity Center (BCCC)
York University
Toronto, ON, Canada
ahabibil@yorku.ca

3rd Ali A. Ghorbani
Canadian Institute for Cybersecurity (CIC)
University of New Brunswick
Fredericton, NB, Canada
ghorbani@unb.ca

2024 21st Annual International Conference on Privacy, Security and Trust (PST) | 979-8-3503-6709-6/24/$31.00 ©2024 IEEE | DOI: 10.1109/PST62714.2024.10788064

***Abstract*—Given their crucial role in protecting networks from numerous security threats, intrusion detection systems are crucial to any cybersecurity architecture. Deep neural networks have recently shown astounding effectiveness and performance in various machine learning applications, including intrusion detection. However, it has been observed that deep learning models are highly susceptible to a wide range of attacks during both the training and testing phases. These attacks can compromise the privacy of deep learning models, such as poisoning attacks that can affect the performance of the target model during the training process and evasion attacks that can undermine the security of these models during the testing phase. Numerous studies have been conducted to understand and mitigate these attacks and to propose more efficient techniques with higher success rates and accuracy in various tasks utilizing deep learning models, such as image classification, face recognition, network intrusion detection, and healthcare applications. Despite the considerable efforts in this area, the network domain still lacks sufficient attention to these attacks and vulnerabilities. This paper aims to address this gap by proposing a framework for adversarial attacks against network intrusion detection systems (NIDS). The proposed framework focuses on poisoning and evasion attacks and tries to combine these attacks. We evaluate the proposed framework on three CIC-IDS2017, CIC-IDS2018, and CIC-UNSW-NB15 datasets.**

***Index Terms*—Network Intrusion Detection, Network Traffic Classification, Deep Learning, Poisoning Attack, Evasion Attack**

I. Introduction

The use of machine learning for automated tasks and decision-making issues has significantly increased recently. ML applications have experienced tremendous growth and are now heavily relied upon in national critical infrastructures and vital sectors such as medicine, healthcare, computer security, spam and malware detection, autonomous driving vehicles, unmanned autonomous systems, and homeland security [1]. These machine-learning models have demonstrated the ability to achieve human-level or even above human-level accuracy in the tasks above. As a result, they are gradually replacing humans or assisting them in carrying out essential duties and making decisions that profoundly impact human lives. Since the emergence of deep learning, these models have excelled at many machine learning tasks and garnered significant attention. Consequently, the security of machine learning and deep learning models has become one of the most intriguing research areas in recent years.

There are different kinds of attacks against deep learning models, which can occur both during the training and testing phases [2], such as membership inference attack [3], [4], model inversion attack [5], evasion attack [6], and poisoning attack [7]. In a membership inference attack, the attacker aims to compromise the model's privacy and determine if a specific record is part of the model's training set. These attacks are particularly significant when dealing with sensitive data, such as health or financial-related information of individuals [8], [9]. The model inversion attack closely resembles the membership inference attack, as it aims to extract sensitive features of the training samples [10]. On the other hand, an evasion attack occurs during the test phase. Its objective is to deceive the trained deep learning model and evade detection by crafting adversarial examples through minor modifications to the original dataset samples [11]. Furthermore, attackers can also impact the model's performance during the training phase, using methods like poisoning attacks in which they try to inject contaminated data into the training set [12].

In recent studies, deep learning has shown considerable promise in improving security measures, especially in fields like malware detection and intrusion detection systems (NIDS). The primary goal of NIDS is to identify and differentiate between attacks directed at a network and benign activities [13]. NIDS operates by continuously analyzing both incoming and outgoing network traffic.

As previously mentioned, deep learning models in the network domain are susceptible to various attacks, with poisoning and evasion attacks being the two major categories [14]. In a poisoning attack, the attacker's objective is to manipulate the training data to compromise the performance of the model. By injecting malicious or misleading data into the training set, the attacker aims to distort the model's learning process, leading to incorrect or biased results during inference.

On the other hand, an evasion attack occurs during the test phase. In this type of attack, the attacker tries to deceive the deep neural network (DNN) model by making subtle changes to the input samples. These changes are designed to evade the model's detection and classification mechanisms, potentially

Authorized licensed use limited to: Telecom SudParis. Downloaded on October 06,2026 at 11:10:31 UTC from IEEE Xplore. Restrictions apply.

causing misclassifications or incorrect outputs [15].

While adversarial attacks on deep learning models in computer vision and image processing have been extensively studied and gained popularity in recent years, the research on such attacks in the network domain has received more attention recently. There have been more attempts to conduct poisoning and evasion attacks against NIDS.

This paper evaluates attacks against Network Intrusion Detection Systems and proposes a comprehensive adversarial attack framework for deep learning-based NIDS. The framework comprises an evasion attack based on a saliency map and a poisoning attack based on label flipping. For the poisoning attack, stratified sampling is employed to select the target label for label flipping. On the other hand, the evasion attack uses a technique to identify the best features for generating adversarial samples for network traffic data. To evaluate the proposed framework, we use CIC-IDS2017, CIC-IDS2018, and CIC-UNSW-NB15 datasets. We first perform a label-flipping attack on the target model during the training phase and then try to generate adversarial samples during the test phase for the poisoned target model. The contributions of this paper are:

- Proposing an adversarial attack framework for deep learning-based NIDS, including poisoning and evasion attacks.
- Evaluating the proposed framework using three well-known datasets, which include several complex and up-to-date network attacks.
- Performing extensive analysis of the results considering several factors.

The rest of the paper first reviews the background and related works. In section three the framework and its components are described the experimental results are shown in section four. Finally, in section five, we have our conclusion and future works.

II. Background and Related Work

In this section, first, we provide a brief introduction to network intrusion detection systems. Then we review the basic concepts of the deep neural network and common attacks against machine learning and deep learning models. After that, the works focusing on poisoning and adversarial cyber attacks are reviewed.

A. Network Intrusion Detection Systems (NIDS)

With the rapid expansion of computer networks and the increasing reliance on digital communication, the security of these networks has become a critical concern. Network intrusion detection systems (NIDS) safeguard computer networks against unauthorized access, malicious activities, and potential cyber threats. NIDS has evolved as a fundamental component of modern network security architectures, providing organizations with a proactive approach to identifying and responding to network attacks [16]. Network intrusion detection systems are designed to monitor network traffic, analyze data packets, and detect suspicious or malicious activities [17]. These systems operate on the principle of analyzing network traffic patterns and comparing them against known attack signatures or abnormal behavior. The NIDS monitors any incoming or departing traffic at the network's edge. The NIDS alerts the public of any suspicious or malicious behavior. NIDS can operate in either a passive or active mode.

Two main categories of NIDS based on detection techniques are signature-based and anomaly-based [18]. Signature-based NIDS, or misuse detection systems, rely on a database of known attack signatures to identify and match patterns within network traffic. They compare incoming traffic against a set of predefined signatures to determine if an attack is in progress. While signature-based NIDS are effective against known attacks and have a low false positive detection rate, they may struggle to detect new or zero-day threats [19], [20]. Anomaly-based NIDS, on the other hand, focuses on establishing a baseline of normal network behavior and flagging any deviations from this baseline as potential intrusions. These systems use machine learning algorithms and statistical analysis to identify anomalies in network traffic [21], [22].

B. Deep Neural Network (DNN)

Several complex real-world problems are difficult to model mathematically and solve even with the significant capabilities of computers. However, the human brain can easily understand these problems, such as speech recognition, image classification, and understanding natural languages. Researchers have long tried to model the human brain's behavior to solve these problems. Finally, in 1985, Frank Rosenblatt introduced perceptron as an artificial model of the human brain's biological neuron [23], [24]. Perceptron can only classify the linearly separable data and cannot solve some simple problems like XOR [25].

Perceptron neurons are arranged in layers to form the Multilayer Perceptron (MLP) [26], which is the foundation of all Deep Neural Networks (DNN). MLP seeks to identify a nonlinear mapping between the attributes of its input layer and each output. Each neuron has a basic function, followed by an activation function like the Sigmoid, TanH, or ReLU. The function of the neuron is to compute the weighted sum of its parameters, which are the outputs of the preceding layer, and then to use the activation function to produce the final outputs of its layer. The output of a neuron in layer j is calculated using the following formula:

$$y_j = f_i(\sum_{i=1}^{n} \theta_{ji} x_i + b_j), \tag{1}$$

where θ_j is the layer j weight matrix, b_j is the bias, and $f(.)$ is the activation function.

The input, hidden, and output layers are the three types of layers that make up an MLP. Depending on the task, each MLP has one input layer, an output layer, and several hidden layers. An MLP model can be considered as the following function F:

Authorized licensed use limited to: Telecom SudParis. Downloaded on October 06,2026 at 11:10:31 UTC from IEEE Xplore. Restrictions apply.

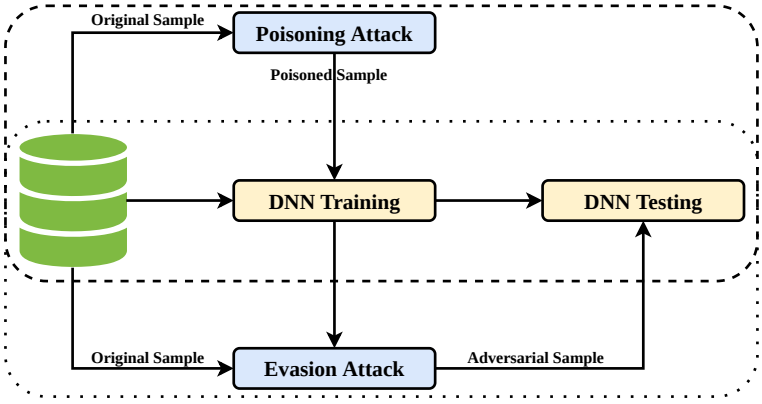


Fig. 1: Overall overview of the proposed framework.

$$F(\vec{x}) = f_n(\theta_n, f_{n-1}(\theta_{n-1}, ...f_2(\theta_2, f_1(\theta_1, \vec{x})))), \quad (2)$$

where θ_i is layer i weight vector, f_i is layer i activation function, and $\vec{x}$ is the input vector.

There are two phases in using a Deep Neural Network (DNN): training and testing. First, a large collection of input-output pairs $(\vec{x}, \vec{y})$ is presented to the initial model to do the training. The model then fine-tunes its weights and biases by reducing the difference between the predicted output and $\vec{y}$. This fine-tuning process commonly employs the back-propagation algorithm [27], [28], coupled with optimization techniques like Stochastic Gradient Descent (SGD) and the Adam optimizer [29]. The trained DNN forecasts outputs for new, unseen inputs in the subsequent testing phase. It relies on the weights and biases computed in the previous phase to carry out these predictions.

C. Attack against Machine Learning and Deep Learning Models

As we mentioned before, ML and DL models are vulnerable to a range of different attacks that can threaten their privacy and security. Some of these attacks are membership inference attack [3], [30]–[33], model inversion attack [5], [34], model extraction attack [35], hyper-parameter stealing attack [36], [37], property inference attack [38]–[40], poisoning attack [41], and evasion attack (adversarial attack) [11], [42]–[46].

Among the attacks mentioned above in this paper, our focus is on poisoning and evasion attacks, and we propose a framework using these attacks against NIDS. In a poisoning attack, the attacker aims to decrease the machine learning model performance by interfering with the training phase [12]. In this attack, the main idea is to inject malicious samples into the training set of the target model [47], [48]. Evasion attacks aim to generate inputs capable of deceiving various machine learning methods, leading them to produce incorrect outcomes. These specially crafted inputs are referred to as adversarial examples. These examples are carefully designed by introducing minor, often undetectable, changes to legitimate inputs, aiming to mislead deep learning models into arriving at inaccurate decisions. A human observer can still accurately classify these examples despite their manipulations [11], [45].

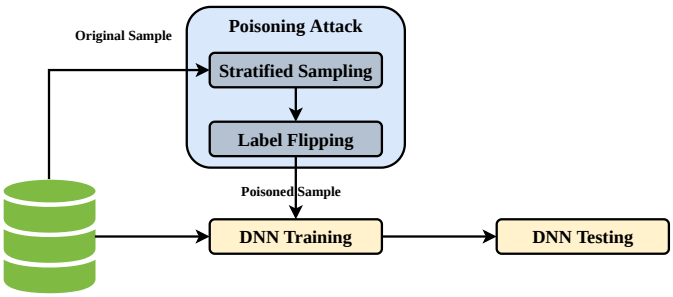


Fig. 2: Label flipping attack.

III. Proposed Method

Figure 1 shows the overall overview of the proposed framework, which consists of two main components: poisoning attack and adversarial attack. The adversarial attack uses the trained model and original samples from the dataset to generate adversarial examples for fooling the deep learning model. However, the poisoning attack happens before the training phase by changing the training set.

For the poisoning attack component, we use the label-flipping attack presented in [49]. In the label-flipping attack, the attacker's goal is to change the labels of the samples in the training set to decrease the performance of the deep learning model. Figure 2 shows the overview of the label-flipping attack, which includes two main steps. Stratified sampling is used to select a subset of the training set as the target records for the label-flipping. Then, the labels of the selected samples are randomly changed to any other label.

For the adversarial attack component, we will use the technique proposed in [50] to generate adversarial samples, changing the fewest possible features. Figure 3 shows the overall overview of this attack. As shown in Figure 3, the proposed method consists of different steps. First, all the possible combinations of the features are extracted, and the

Authorized licensed use limited to: Telecom SudParis. Downloaded on October 06,2026 at 11:10:31 UTC from IEEE Xplore. Restrictions apply.

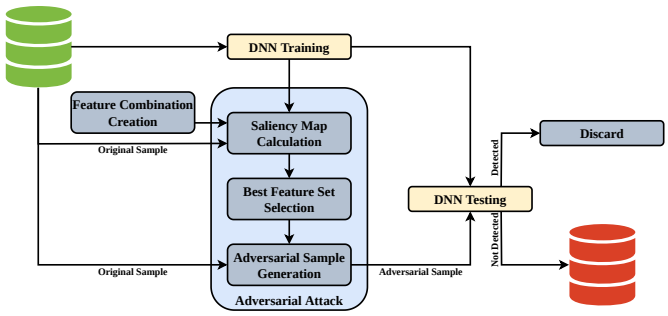


Fig. 3: Saliency map-based attack.

saliency map is calculated for these combinations. Then, the feature combination that has the highest value will be used to generate the adversarial example using the following equation:

$$X' = X + mask * \epsilon. \tag{3}$$

X and X' are the original and adversarial samples, $mask$ is the selected feature combination from previous step, and ϵ is the value added to each feature.

The proposed framework has two components: poisoning attack and evasion attack. The target model is trained under a label-flipping attack, and the evasion attack is performed.

IV. Experiments and Results

In this Section, we first provide the details of the datasets we used in the experiments. Then the results of our experiments showing the target model under both label-flipping and saliency map-based attacks.

A. Dataset

One of the critical factors in researching NIDS is selecting a comprehensive dataset that includes an extensive set of labeled intrusions and abnormal behavior that covers up-to-date network attacks. To do our experiments, we decided on three datasets. CIC-IDS2017 [51] and CIC-IDS2018, which are from our institute. To have more evaluation on an external dataset, we chose UNSW-NB15 [52], [53].

CIC-IDS2017 [51] and CIC-IDS2018 datasets include various instances of network attacks. In CIC-IDS2017, they tried to cover all the necessary criteria mentioned in [54] for an IDS dataset. CIC-IDS2018 is an extension of CIC-IDS2017, which has the same attacks. CICFlowMeter [55] extracts network flows with more than 80 features. The details of these two datasets can be found in Table I and Table II. Some features, such as Flow ID, Source IP, Source Port, Destination IP, Destination Port, Protocol, and Timestamp, are not useful for training a DNN model. We removed them during the prepossessing step.

To generate the UNSW-NB15 dataset, they used the IXIA PerfectStorm tool to create modern normal and abnormal network traffic. Their dataset includes nine attack categories and benign traffic. They captured 100GBs of network traffic in two days, and to extract features from the captured network traffic, they used Argus and Bro-IDS tools. They extracted 47 features in categories, including Basic, Content, Time, and additional generated features.

TABLE I: CIC-IDS2017 details.

Attack Name	Number of Records
Benign	2271320
DDoS	128025
PortScan	158804
Botnet	1956
Infiltration	36
Web Attack-Brute Force	1507
Web Attack-Sql Injection	21
Web Attack-XSS	652
FTP-Patator	7935
SSH-Patator	5897
DoS GoldenEye	10293
DoS Hulk	230124
DoS Slowhttp	5499
Dos Slowloris	5796
Heartbleed	11

TABLE II: CIC-IDS2018 details.

Attack Name	Number of Records
Benign	13390249
Bot	286191
Infiltration	160639
Brute Force-Web	611
Brute Force-XSS	230
SQL Injection	87
FTP-BruteForce	193354
SSH-BruteForce	187589
DoS GoldenEye	41508
DoS Hulk	461912
DoS Slowhttp	139890
Dos Slowloris	10990
DDoS LOIC-HTTP	576191
DDoS LOIC-UDP	1730
DDoS HOIC	686012

Since they have provided all the pcap files and we require to have datasets with the same set of features, we used CICFlowMeter to extract the new set of features from the provided captured network traffics. After extracting the flows using CICFlowMeter we need to label them using the ground truth provided with the original dataset files.

We matched the extracted flows with the records in the ground truth file based on the source IP, destination IP, source port, destination port, and protocol. If any flows match with a record from the ground truth file, we set the label using the ground truth attack category. If the flow is matched with more than one record from the ground truth file, we compare the timestamps and choose the label of the record that matches the flow timestamp. In the worst case, even if by comparing the timestamp we can not decide the label, the flow will be dropped. Any remaining flows will be labeled benign after labeling all the malicious flows.

Table IV includes the details of the original UNSW-NB15 dataset and the extracted flows using CICFlowMeter. In most network traffic datasets, in order to be closer to the real world, they keep the ratio between benign and malicious samples from 80 percent to 20 percent. To gain this ratio, we

Authorized licensed use limited to: Telecom SudParis. Downloaded on October 06,2026 at 11:10:31 UTC from IEEE Xplore. Restrictions apply.

TABLE III: Results of the Classifiers

Machine Learning Techniques	CIC-IDS2017			CIC-IDS2018			CIC-UNSW-NB15		
	F1-score	PC	RC	F1-score	PC	RC	F1-score	PC	RC
DT	99.84	99.76	99.92	97.70	99.73	96.14	98.81	98.93	98.7
Naive Bayes	28.47	32.43	73.75	48.29	49.48	72.79	22.43	43.14	44.57
LR	36.71	39.76	34.96	57.97	64.13	56.84	32.99	46.26	33.61
RF	96.58	99.79	94.24	94.23	99.81	90.97	91.87	96.06	88.79
DNN	98.18	98.27	98.22	98.71	98.99	99.04	92.06	93.63	92.14

TABLE IV: CIC-UNSW-NB15 details.

Category	Original Dataset	CICFlowMeter	CIC-UNSW-NB15
Benign	221876	3450658	358332
Analysis	2677	385	385
Backdoor	2329	452	452
DoS	16353	4467	4467
Exploits	44525	30951	30951
Fuzzers	24246	29613	29613
Generic	215481	4632	4632
Reconnaissance	13987	16735	16735
Shellcode	1511	2102	2102
Worms	174	246	246

keep all the malicious flows extracted by the CICFlowMeter and randomly sample the required number of flows from the benign flows. The last column of Table IV shows the final details of the newly generated dataset. Since we used the raw packet files from UNSW-NB15 and the CICFlowMeter to generate this dataset and augment UNSW-NB15, we will call it CIC-UNSW-NB15. The CIC-UNSW-NB15 dataset has been published on CIC's dataset page, making it available for other researchers to use[1].

B. Target Model Training

Since the proposed framework is for attacks against deep learning-based NIDS, we start by training a target model and comparing its performance with other traditional machine learning models. We train a multi-layer perceptron with two hidden layers; each of them contains 256 neurons. During the training, the inputs are 76 features of each network flow, and the outputs are different probability values for each respective attack type and benign flow. We used RelU as our activation function and a Dropout layer with 0.2 probability in both hidden layers. The results provided in Table III show that our model has comparable performance with other machine learning models.

C. Results

As explained before we are going to perform the saliency map-based attack against a target model that has been under label flipping attack during its training phase. In our experiments, we did the label-flipping attack with 10, 20, and 30 percent flipped labels and performed the saliency map-based attack with one, two, or three features and ϵ values 0.1, 0.01, and 0.001. In the following subsections, we will present the combined attack results for each dataset separately.

[1]https://www.unb.ca/cic/datasets/cic-unsw-nb15.html

1) CIC-IDS2017: In Table V, the results of the combined attack for CIC-IDS2017 are presented. In general, when the target model is under a label flipping attack, the success rate of the adversarial attack clearly drops even with 10 percent flipped labels. Also, in almost all cases, increasing the number of features while keeping the ϵ value unchanged or increasing the ϵ with the same number of features increases the success rate. With the same value of ϵ and the number of features, increasing the flipped labels percentage does not always have the same effect. With $\epsilon = 0.01$ and three features increasing, the flip percentage increases the attack's success, but with $\epsilon = 0.1$ and two features, the success rate first increased and then decreased.

The top feature groups with different ϵ values and the amount of flipped labels are shown in Table VI. In the presence of a label-flipping attack, it is not possible to make a general assumption about the most successful feature groups, even in the same label-flipping attack. Introducing more flipped labels during the training phase improves the effect of some features on adversarial sample generation. With 10% flipped, *Fwd Act Data Pkts* is present in two top feature groups, with 20% flipped in five top feature groups, and finally in all the cases with 30% flip. The same thing happens with the combination of *Total Fwd Packet, Subflow Fwd Packets*, and *Fwd Act Data Pkts* where five of the top feature groups with 30% flip are this combination.

2) CIC-IDS2018: The result of the combined attack for CIC-IDS2018 is completely different from CIC-IDS2017. Here, the adversarial attack performance is always better than when there is no label-flipping attack. As expected, increasing the number of features or the ϵ value increases the saliency map-based attack success rate. Increasing the number of flipped labels does not have an understandable relation with the number of generated adversarial samples. In some cases, the success rate keeps increasing; in others, it decreases and then increases. Table VII shows the detailed results of CIC-IDS2018.

The top three best feature groups of the combined attack for CIC-IDS2018 are shown in Table VIII. Unlike CIC-IDS2017, there are many similarities between the top feature groups for CIC-IDS2018. For the attack on the target model with 10% flipped labels, the top three feature groups for different ϵ values are the same but with different rankings. For the 20% flip, the feature groups are also the same except for $\epsilon = 0.001$, where one of the features of the third best group is different. With ϵ values of 0.1 and 0.01 for the 30% flipped attack, the top features are the same. But with $\epsilon = 0.001$, the feature groups

Authorized licensed use limited to: Telecom SudParis. Downloaded on October 06,2026 at 11:10:31 UTC from IEEE Xplore. Restrictions apply.

TABLE V: Combined attack results for CIC-IDS2017.

Flip %	10				20				30			
ϵ	All	1 Feature	2 Features	3 Features	All	1 Feature	2 Features	3 Features	All	1 Feature	2 Features	3 Features
0.1	629871	308274	385399	370666	504644	260303	398862	365101	648208	275192	334839	394709
	22.69	**11.1**	**13.88**	**13.35**	**18.28**	**9.43**	**14.44**	**13.22**	**23.51**	**9.98**	**12.14**	**14.31**
0.01	474727	114089	176824	181575	479248	147531	184202	184580	544936	174196	232044	224843
	17.1	**4.11**	**6.37**	**6.54**	**17.36**	**5.34**	**6.67**	**6.68**	**19.76**	**6.31**	**8.41**	**8.15**
0.001	211920	114089	43789	53935	219441	17872	37490	70504	188737	7311	16772	55185
	7.63	**1.29**	**1.57**	**1.94**	**7.94**	**0.64**	**1.35**	**2.55**	**7.21**	**0.26**	**0.6**	**2**

TABLE VI: CIC-IDS2017 top 3 best feature sets for the combined attack.

10					
0.1		**0.01**		**0.001**	
Samples	**Selected Features**	**Samples**	**Selected Features**	**Samples**	**Selected Features**
77828	Fwd Act Data Pkts	28831	Flow IAT Min	22106	Flow IAT Min
64282	Subflow Fwd Packets	27374	Total Bwd packets Flow IAT Min Subflow Bwd Packets	20595	Flow IAT Mean Flow IAT Min
62482	Total Fwd Packets Subflow Fwd Packets	27342	Fwd Act Data Pkts	8880	Flow IAT Mean Flow IAT Min Fwd IAT Min
20					
0.1		**0.01**		**0.001**	
Samples	**Selected Features**	**Samples**	**Selected Features**	**Samples**	**Selected Features**
85300	Fwd Act Data Pkts	79286	Fwd Act Data Pkts	20683	Flow Duration Flow IAT Max Fwd Act Data Pkts
61391	Total Bwd packets Subflow Bwd Packets Fwd Act Data Pkts	62173	Total Bwd packets Subflow Bwd Packets Fwd Act Data Pkts	15033	Packet Length Min Down/Up Ratio
45371	Total Bwd packets	42863	Total Bwd packets Subflow Bwd Packets	14925	Packet Length Min Down/Up Ratio FWD Init Win Bytes
30					
0.1		**0.01**		**0.001**	
Samples	**Selected Features**	**Samples**	**Selected Features**	**Samples**	**Selected Features**
148382	Fwd Act Data Pkts	85271	Fwd Act Data Pkts	22167	Total Bwd packets Subflow Bwd Packets Fwd Act Data Pkts
101123	Total Bwd packets Subflow Bwd Packets Fwd Act Data Pkts	78135	Total Fwd Packet Subflow Fwd Packets Fwd Act Data Pkts	12964	Total Bwd packets Bwd Packet Length Max Fwd Act Data Pkts
86026	Total Bwd packets Fwd Act Data Pkts	77797	Total Bwd packets Subflow Bwd Packets Fwd Act Data Pkts	8378	Total Bwd packets Fwd Act Data Pkts

are the same as the 10% flipped attack.

3) CIC-UNSW-NB15: We presented the results for CIC-UNSW-NB15 in Table IX. Other than the results with the 10% flipped attack and $\epsilon = 0.1$, all the success rates are worse than the original saliency map-based attack. It seems, in general, that the attack on the target model with more flipped labels has less success, and the number of generated samples is smaller. Also, the same as the two other datasets, increasing the number of features or the ϵ value improves the attack's performance.

The combined attack's top feature groups are completely different from the original saliency map-based attack. Table X presents a detailed list of best features. As with the CIC-IDS2017, there is not much similarity between the top feature groups for the CIC-UNSW-NB15, with different percentages of the flipped labels. Some features like *PSH Flag Count*, *Down/Up Ratio*, and *Bwd IAT Min* look more effective than others. But, a general decision can not be made based on the presented findings.

V. Conclusion and Future Works

Intrusion detection systems play a crucial role in any cybersecurity infrastructure. However, with the rapid increase in network traffic speed and density, traditional intrusion detection systems need help efficiently detect sophisticated attacks. In recent years, deep neural networks have shown remarkable performance and efficiency in various machine learning tasks, including intrusion detection. Nevertheless, researchers have discovered that machine learning and deep learning models are susceptible to attacks compromising security and privacy. Some of these attacks occur during the training phase, such as poisoning attacks, while others exploit vulnerabilities in trained models, including membership inference attacks, model inversion attacks, and evasion attacks.

Despite these attacks' significance, most existing research has focused on domains other than network security, leaving a gap in understanding their impact on network-related tasks. Furthermore, studies in the network domain have often borrowed attack techniques from unrelated disciplines without customizing them to suit the unique characteristics of network data. Moreover, evaluations have been conducted using more complex datasets. Therefore, there is a pressing need for research specifically tailored to the network domain, considering the peculiarities of network data and employing state-of-the-art attack techniques. Such efforts will help bolster the security

Authorized licensed use limited to: Telecom SudParis. Downloaded on October 06,2026 at 11:10:31 UTC from IEEE Xplore. Restrictions apply.

TABLE VII: Combined attack results for CIC-IDS2018.

Flip %	10				20				30			
ϵ	All	1 Feature	2 Features	3 Features	All	1 Feature	2 Features	3 Features	All	1 Feature	2 Features	3 Features
0.1	2451237	1687428	2201309	2164226	2941030	2219836	2510622	2985471	3081374	1337511	2367677	2483340
	15.49	**10.66**	**13.91**	**13.68**	**18.64**	**14.06**	**15.91**	**18.92**	**19.53**	**8.47**	**15**	**15.74**
0.01	2842554	998248	1855722	1794083	2974524	981066	1484063	1794128	3007534	889846	1327110	1931678
	17.97	**6.31**	**11.73**	**11.34**	**18.85**	**6.21**	**9.4**	**11.37**	**19.06**	**5.64**	**8.41**	**12.24**
0.001	2153134	577312	813897	856811	1611037	282524	324749	272954	1633541	266534	177377	304872
	13.61	**3.65**	**5.14**	**5.41**	**10.21**	**1.79**	**2.05**	**1.73**	**10.35**	**1.68**	**1.12**	**1.93**

TABLE VIII: CIC-IDS2018 top 3 best feature sets for the combined attack.

10					
0.1		**0.01**		**0.001**	
Samples	**Selected Features**	**Samples**	**Selected Features**	**Samples**	**Selected Features**
552409	Idle Min	537715	Idle Mean Idle Max Idle Min	536706	Idle Mean Idle Max Idle Min
527001	Idle Mean Idle Max Idle Min	535132	Idle Min	530229	Idle Mean Idle Min
523484	Idle Mean Idle Min	530820	Idle Mean Idle Min	523378	Idle Min
20					
0.1		**0.01**		**0.001**	
Samples	**Selected Features**	**Samples**	**Selected Features**	**Samples**	**Selected Features**
1215432	Fwd Packets/s	1053043	Flow Packets/s Fwd Packets/s	236450	Flow Packets/s Fwd Packets/s
1214704	Flow Packets/s Fwd Packets/s	771014	Fwd Packets/s	233273	Fwd Packets/s
685659	Flow Packets/s Fwd Packets/s Bwd Packets/s	521909	Flow Packets/s Fwd Packets/s Bwd Packets/s	80758	Flow Packets/s Fwd Packets/s Bwd Init Win Bytes
30					
0.1		**0.01**		**0.001**	
Samples	**Selected Features**	**Samples**	**Selected Features**	**Samples**	**Selected Features**
721033	Fwd Packets/s	699030	Fwd Packets/s	126410	Idle Mean Idle Max Idle Min
685774	Fwd Packet Length Std Flow Packets/s Fwd Packets/s	685528	Fwd Packet Length Std Flow Packets/s Fwd Packets/s	123397	Idle Min
685774	Fwd Packets/s Fwd Segment Size Avg	685524	Fwd Packets/s Fwd Segment Size Avg	122805	Idle Mean Idle Min

of intrusion detection systems and fortify overall cybersecurity measures.

This paper presented a comprehensive framework comprising two primary components: poisoning and evasion attacks targeting deep learning-based Network Intrusion Detection Systems (NIDS). We used an approach based on label-flipping for the poisoning attack and for the evasion attack, a technique based on saliency maps. The analysis of the combined attack reveals some similarities among the three datasets, but drawing a general conclusion is challenging. When an evasion attack is executed on a model trained on a poisoned dataset, several factors come into play, affecting the attack's success rate.

One crucial factor is the reduced accuracy of the target model due to the poisoning attack. As the target model's accuracy decreases, the number of detected samples available for generating adversarial examples also reduces, potentially impacting the success rate of the evasion attack. On the other hand, the decreased accuracy of the target model might make it more susceptible to being fooled by adversarial samples. This could work in favor of the evasion attack, increasing its success rate.

With these two opposing effects at play, the results of the combined attack become inconclusive. In some instances, increasing the percentage of flipped labels in the poisoning attack might lead to a higher success rate for the evasion attack. However, in other cases, it may not have a significant impact on the success rate.

Instead of label flipping, poisoning attacks offer alternative techniques like data injection and poisoning that researchers should consider for future investigations. The majority of existing works on evasion attacks are confined to the feature space, which is susceptible to changes in the nature of the flow. In the future, our focus in the feature space should be on ensuring that attackers' modifications to the flow's features do not alter the nature of that flow. Moreover, an alternative direction worth exploring is conducting deep packet adversarial attacks instead of restricting ourselves to the feature space. Furthermore, we have focused solely on attacks against deep learning-based NIDS without providing any defense mechanisms against such attacks. To address this crucial gap, a promising future direction is to delve into existing defense methods designed to counter these attacks. By exploring and studying these defense techniques, we can gain insights into their effectiveness and limitations.

Authorized licensed use limited to: Telecom SudParis. Downloaded on October 06,2026 at 11:10:31 UTC from IEEE Xplore. Restrictions apply.

TABLE IX: Combined attack results for CIC-UNSW-NB15.

Flip %	10				20				30			
ϵ	All	1 Feature	2 Features	3 Features	All	1 Feature	2 Features	3 Features	All	1 Feature	2 Features	3 Features
0.1	133776	34099	73896	99216	163824	37850	46250	51945	123383	35859	45725	53287
	32.53	**8.29**	**17.97**	**24.13**	**39.88**	**9.21**	**11.25**	**12.64**	**30.06**	**8.73**	**11.14**	**12.98**
0.01	43585	5965	14502	17759	40757	3270	8974	17748	43435	5166	9888	11896
	10.6	**1.45**	**3.52**	**4.31**	**9.92**	**0.79**	**2.18**	**4.32**	**10.58**	**1.25**	**2.4**	**2.89**
0.001	12271	192	355	590	8648	294	551	779	7984	202	348	488
	2.98	**0.04**	**0.08**	**0.14**	**2.1**	**0.07**	**0.13**	**0.18**	**1.94**	**0.04**	**0.08**	**0.11**

TABLE X: CIC-UNSW-NB15 top 3 best feature sets for the combined attack.

10					
0.1		**0.01**		**0.001**	
Samples	**Selected Features**	**Samples**	**Selected Features**	**Samples**	**Selected Features**
37248	Total Bwd packets Fwd IAT Min Bwd IAT Total	3395	PSH Flag Count Down/Up Ratio	119	Total Bwd packets Bwd IAT Total Bwd Header Length
29595	Total Bwd packets Bwd IAT Total	3270	Packet Length Variance Bwd Packet/Bulk Avg	57	Flow IAT Std Fwd IAT Mean PSH Flag Count
9948	Flow IAT Min Fwd IAT Min Bwd IAT Min	2627	Fwd IAT Total PSH Flag Count Bwd Packet/Bulk Avg	49	Total Bwd packets Bwd Header Length
20					
0.1		**0.01**		**0.001**	
Samples	**Selected Features**	**Samples**	**Selected Features**	**Samples**	**Selected Features**
11103	Bwd IAT Min	3566	PSH Flag Count Down/Up Ratio	181	Total Bwd packets Bwd IAT Total Bwd Header Length
10191	PSH Flag Count Down/Up Ratio	3259	Packet Length Variance Bwd Segment Size Avg Bwd Packet/Bulk Avg	92	Total Bwd packets Bwd IAT Total
8026	Bwd IAT Min SYN Flag Count	2987	Fwd IAT Total Fwd IAT Mean Fwd IAT Std	91	PSH Flag Count Down/Up Ratio Subflow Fwd Packets
30					
0.1		**0.01**		**0.001**	
Samples	**Selected Features**	**Samples**	**Selected Features**	**Samples**	**Selected Features**
11809	PSH Flag Count Down/Up Ratio	3257	Total Length of Bwd Packet PSH Flag Count Down/Up Ratio	85	Flow IAT Std Flow IAT Max
11057	Down/Up Ratio	3097	PSH Flag Count Down/Up Ratio	66	Flow IAT Mean Flow IAT Std Flow IAT Max
9950	Bwd IAT Min	2971	Packet Length Variance Bwd Packet/Bulk Avg	60	Flow IAT Std

REFERENCES

[1] V. Duddu, "A survey of adversarial machine learning in cyber warfare.," *Defence Science Journal*, vol. 68, no. 4, 2018.

[2] B. Biggio and F. Roli, "Wild patterns: Ten years after the rise of adversarial machine learning," *Pattern Recognition*, vol. 84, pp. 317–331, 2018.

[3] R. Shokri, M. Stronati, C. Song, and V. Shmatikov, "Membership inference attacks against machine learning models," in *2017 IEEE symposium on security and privacy (SP)*, pp. 3–18, IEEE, 2017.

[4] H. Hu, Z. Salcic, L. Sun, G. Dobbie, P. S. Yu, and X. Zhang, "Membership inference attacks on machine learning: A survey," *ACM Computing Surveys (CSUR)*, vol. 54, no. 11s, pp. 1–37, 2022.

[5] M. Fredrikson, S. Jha, and T. Ristenpart, "Model inversion attacks that exploit confidence information and basic countermeasures," in *Proceedings of the 22nd ACM SIGSAC conference on computer and communications security*, pp. 1322–1333, 2015.

[6] C. Szegedy, W. Zaremba, I. Sutskever, J. Bruna, D. Erhan, I. Goodfellow, and R. Fergus, "Intriguing properties of neural networks," *arXiv preprint arXiv:1312.6199*, 2013.

[7] Z. Tian, L. Cui, J. Liang, and S. Yu, "A comprehensive survey on poisoning attacks and countermeasures in machine learning," *ACM Computing Surveys*, vol. 55, no. 8, pp. 1–35, 2022.

[8] S. Truex, L. Liu, M. E. Gursoy, L. Yu, and W. Wei, "Towards demystifying membership inference attacks," *arXiv preprint arXiv:1807.09173*, 2018.

[9] C. A. Choquette-Choo, F. Tramer, N. Carlini, and N. Papernot, "Label-only membership inference attacks," in *International conference on machine learning*, pp. 1964–1974, PMLR, 2021.

[10] Y. Zhang, R. Jia, H. Pei, W. Wang, B. Li, and D. Song, "The secret revealer: Generative model-inversion attacks against deep neural networks," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 253–261, 2020.

[11] I. J. Goodfellow, J. Shlens, and C. Szegedy, "Explaining and harnessing adversarial examples," *arXiv preprint arXiv:1412.6572*, 2014.

[12] Z. Wang, J. Ma, X. Wang, J. Hu, Z. Qin, and K. Ren, "Threats to training: A survey of poisoning attacks and defenses on machine learning systems," *ACM Computing Surveys*, vol. 55, no. 7, pp. 1–36, 2022.

[13] A. L. Buczak and E. Guven, "A survey of data mining and machine learning methods for cyber security intrusion detection," *IEEE Communications surveys & tutorials*, vol. 18, no. 2, pp. 1153–1176, 2015.

[14] Y. Peng, J. Su, X. Shi, and B. Zhao, "Evaluating deep learning based network intrusion detection system in adversarial environment," in *2019 IEEE 9th International Conference on Electronics Information and Emergency Communication (ICEIEC)*, pp. 61–66, IEEE, 2019.

[15] N. Pitropakis, E. Panaousis, T. Giannetsos, E. Anastasiadis, and G. Loukas, "A taxonomy and survey of attacks against machine learning," *Computer Science Review*, vol. 34, p. 100199, 2019.

[16] B. Mukherjee, L. T. Heberlein, and K. N. Levitt, "Network intrusion detection," *IEEE network*, vol. 8, no. 3, pp. 26–41, 1994.

[17] S. K. Jonnalagadda and R. P. Reddy, "A literature survey and comprehensive study of intrusion detection," *International Journal of Computer Applications*, vol. 81, no. 16, pp. 40–47, 2013.

[18] E. Hodo, X. Bellekens, A. Hamilton, C. Tachtatzis, and R. Atkinson,

Authorized licensed use limited to: Telecom SudParis. Downloaded on October 06,2026 at 11:10:31 UTC from IEEE Xplore. Restrictions apply.

"Shallow and deep networks intrusion detection system: A taxonomy and survey," *arXiv preprint arXiv:1701.02145*, 2017.

[19] H. Debar, M. Dacier, and A. Wespi, "Towards a taxonomy of intrusion-detection systems," *Computer networks*, vol. 31, no. 8, pp. 805–822, 1999.

[20] H. E. Poston, "A brief taxonomy of intrusion detection strategies," in *2012 IEEE National Aerospace and Electronics Conference (NAECON)*, pp. 255–263, IEEE, 2012.

[21] N. K. Mittal, "A survey on wireless sensor network for community intrusion detection systems," in *2016 3rd International Conference on Recent Advances in Information Technology (RAIT)*, pp. 107–111, IEEE, 2016.

[22] J. Shun and H. A. Malki, "Network intrusion detection system using neural networks," in *2008 fourth international conference on natural computation*, vol. 5, pp. 242–246, IEEE, 2008.

[23] F. Rosenblatt, "The perceptron: a probabilistic model for information storage and organization in the brain.," *Psychological review*, vol. 65, no. 6, p. 386, 1958.

[24] F. Rosenblatt, "Principles of neurodynamics. perceptrons and the theory of brain mechanisms," tech. rep., Cornell Aeronautical Lab Inc Buffalo NY, 1961.

[25] M. Minsky and S. A. Papert, *Perceptrons: An introduction to computational geometry*. MIT press, 2017.

[26] K. Hornik, M. Stinchcombe, and H. White, "Multilayer feedforward networks are universal approximators," *Neural networks*, vol. 2, no. 5, pp. 359–366, 1989.

[27] S. Dreyfus, "The computational solution of optimal control problems with time lag," *IEEE Transactions on Automatic Control*, vol. 18, no. 4, pp. 383–385, 1973.

[28] P. J. Werbos, "Applications of advances in nonlinear sensitivity analysis," in *System modeling and optimization*, pp. 762–770, Springer, 1982.

[29] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," *arXiv preprint arXiv:1412.6980*, 2014.

[30] S. Hisamoto, M. Post, and K. Duh, "Membership inference attacks on sequence-to-sequence models: Is my data in your machine translation system?," *Transactions of the Association for Computational Linguistics*, vol. 8, pp. 49–63, 2020.

[31] G. Liu, C. Wang, K. Peng, H. Huang, Y. Li, and W. Cheng, "Socinf: Membership inference attacks on social media health data with machine learning," *IEEE Transactions on Computational Social Systems*, vol. 6, no. 5, pp. 907–921, 2019.

[32] A. Salem, Y. Zhang, M. Humbert, P. Berrang, M. Fritz, and M. Backes, "Ml-leaks: Model and data independent membership inference attacks and defenses on machine learning models," *arXiv preprint arXiv:1806.01246*, 2018.

[33] S. Truex, L. Liu, M. E. Gursoy, L. Yu, and W. Wei, "Demystifying membership inference attacks in machine learning as a service," *IEEE Transactions on Services Computing*, vol. 14, no. 6, pp. 2073–2089, 2019.

[34] M. Fredrikson, E. Lantz, S. Jha, S. Lin, D. Page, and T. Ristenpart, "Privacy in pharmacogenetics: An end-to-end case study of personalized warfarin dosing," in *23rd {USENIX} Security Symposium ({USENIX} Security 14)*, pp. 17–32, 2014.

[35] F. Tramèr, F. Zhang, A. Juels, M. K. Reiter, and T. Ristenpart, "Stealing machine learning models via prediction apis.," in *USENIX security symposium*, vol. 16, pp. 601–618, 2016.

[36] S. J. Oh, B. Schiele, and M. Fritz, "Towards reverse-engineering black-box neural networks," *Explainable AI: Interpreting, Explaining and Visualizing Deep Learning*, pp. 121–144, 2019.

[37] B. Wang and N. Z. Gong, "Stealing hyperparameters in machine learning," in *2018 IEEE symposium on security and privacy (SP)*, pp. 36–52, IEEE, 2018.

[38] G. Ateniese, L. V. Mancini, A. Spognardi, A. Villani, D. Vitali, and G. Felici, "Hacking smart machines with smarter ones: How to extract meaningful data from machine learning classifiers," *International Journal of Security and Networks*, vol. 10, no. 3, pp. 137–150, 2015.

[39] K. Ganju, Q. Wang, W. Yang, C. A. Gunter, and N. Borisov, "Property inference attacks on fully connected neural networks using permutation invariant representations," in *Proceedings of the 2018 ACM SIGSAC conference on computer and communications security*, pp. 619–633, 2018.

[40] L. Melis, C. Song, E. De Cristofaro, and V. Shmatikov, "Exploiting unintended feature leakage in collaborative learning," in *2019 IEEE symposium on security and privacy (SP)*, pp. 691–706, IEEE, 2019.

[41] F. Tramèr, A. Kurakin, N. Papernot, I. Goodfellow, D. Boneh, and P. McDaniel, "Ensemble adversarial training: Attacks and defenses," *arXiv preprint arXiv:1705.07204*, 2017.

[42] N. Carlini and D. Wagner, "Towards evaluating the robustness of neural networks," in *2017 ieee symposium on security and privacy (sp)*, pp. 39–57, IEEE, 2017.

[43] B. Li and Y. Vorobeychik, "Scalable optimization of randomized operational decisions in adversarial classification settings," in *Artificial Intelligence and Statistics*, pp. 599–607, PMLR, 2015.

[44] N. Papernot, P. McDaniel, S. Jha, M. Fredrikson, Z. B. Celik, and A. Swami, "The limitations of deep learning in adversarial settings," in *2016 IEEE European symposium on security and privacy (EuroS&P)*, pp. 372–387, IEEE, 2016.

[45] N. Papernot, P. McDaniel, I. Goodfellow, S. Jha, Z. B. Celik, and A. Swami, "Practical black-box attacks against machine learning," in *Proceedings of the 2017 ACM on Asia conference on computer and communications security*, pp. 506–519, 2017.

[46] W. Xu, D. Evans, and Y. Qi, "Feature squeezing: Detecting adversarial examples in deep neural networks," *arXiv preprint arXiv:1704.01155*, 2017.

[47] M. Barreno, B. Nelson, A. D. Joseph, and J. D. Tygar, "The security of machine learning," *Machine Learning*, vol. 81, pp. 121–148, 2010.

[48] M. Barreno, B. Nelson, R. Sears, A. D. Joseph, and J. D. Tygar, "Can machine learning be secure?," in *Proceedings of the 2006 ACM Symposium on Information, computer and communications security*, pp. 16–25, 2006.

[49] H. Mohammadian., A. Lashkari., and A. A. Ghorbani., "Evaluating label flipping attack in deep learning-based nids," in *Proceedings of the 20th International Conference on Security and Cryptography - SECRYPT*, pp. 597–603, INSTICC, SciTePress, 2023.

[50] H. Mohammadian, A. A. Ghorbani, and A. H. Lashkari, "A gradient-based approach for adversarial attack on deep learning-based network intrusion detection systems," *Applied Soft Computing*, vol. 137, p. 110173, 2023.

[51] I. Sharafaldin, A. H. Lashkari, and A. A. Ghorbani, "Toward generating a new intrusion detection dataset and intrusion traffic characterization.," in *ICISSP*, pp. 108–116, 2018.

[52] N. Moustafa and J. Slay, "Unsw-nb15: a comprehensive data set for network intrusion detection systems (unsw-nb15 network data set)," in *2015 military communications and information systems conference (MilCIS)*, pp. 1–6, IEEE, 2015.

[53] N. Moustafa and J. Slay, "The evaluation of network anomaly detection systems: Statistical analysis of the unsw-nb15 data set and the comparison with the kdd99 data set," *Information Security Journal: A Global Perspective*, vol. 25, no. 1-3, pp. 18–31, 2016.

[54] A. Gharib, I. Sharafaldin, A. H. Lashkari, and A. A. Ghorbani, "An evaluation framework for intrusion detection dataset," in *2016 International Conference on Information Science and Security (ICISS)*, pp. 1–6, IEEE, 2016.

[55] A. H. Lashkari, G. Draper-Gil, M. S. I. Mamun, and A. A. Ghorbani, "Characterization of tor traffic using time based features.," in *ICISSp*, pp. 253–262, 2017.

Authorized licensed use limited to: Telecom SudParis. Downloaded on October 06,2026 at 11:10:31 UTC from IEEE Xplore. Restrictions apply.